

Automated search recovers exact rank-7 matrix multiplication

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Abstract

Recent work proving that the matrix-multiplication exponent satisfies $\omega \leq 9/4$ motivates the search for explicit finite algorithms with improved recursive complexity. We present **fast_mm_search**, a numerical-to-exact procedure that searches for bilinear matrix-multiplication decompositions from random coefficients, reconstructs exact identities, and verifies them using exact arithmetic. As a controlled validation, we specified only the 2×2 , rank-7 problem and withheld Strassen’s known construction. The search recovered an exact seven-product algorithm with integer coefficients and a representation different from the standard Strassen formula. Exact rational verification established equality for all 64 coefficients of the multiplication tensor. This blind recovery demonstrates that exact finite matrix-multiplication algorithms can be recovered automatically and provides a validated procedure for searching finite (u, r) decompositions whose explicit solutions are unknown.

1 Introduction

Strassen showed that 2×2 matrix multiplication can be computed with seven bilinear products instead of the eight used by the direct algorithm (1). Recursive application gives arithmetic exponent

$$\log_2 7 \simeq 2.80735.$$

Recent work proves the stronger asymptotic bound

$$\omega \leq \frac{9}{4}$$

over $\mathbb{C}$ (2). This motivates the constructive search for explicit finite matrix-multiplication decompositions with improved recursive complexity.

We developed `fast_mm_search` to search for such decompositions numerically, reconstruct candidate coefficients exactly, and verify the resulting identities independently. The 2×2 , rank-7 problem provides a controlled validation because an exact solution is known while its coefficients are withheld from the search.

2 Methods

Let

$$T \in \mathbb{Q}^{4 \times 4 \times 4}$$

denote the tensor for 2×2 matrix multiplication. A rank- r bilinear algorithm has the decomposition

$$\hat{T} = \sum_{k=1}^r u_k \otimes v_k \otimes w_k,$$

where u_k and v_k define linear forms in the entries of the two input matrices and w_k distributes each bilinear product among the output entries.

We fixed $r = 7$ and optimised the coefficients from independent random initial values using L-BFGS-B. Candidates were ranked by relative tensor residual

$$\rho = \frac{\|T - \hat{T}\|_F}{\|T\|_F}.$$

Candidates reaching the numerical threshold were term-normalised, reconstructed as rational coefficients, and passed to a separate verifier using exact rational arithmetic. Numerical optimisation selects candidates; exact tensor equality determines acceptance.

3 Results

The campaign evaluated 581 independent random initialisations. The lowest numerical residual,

$$1.596061 \times 10^{-8},$$

occurred at seed 482 and did not yield an exact verified identity. Seed 581 converged after 140 L-BFGS-B iterations to

$$\rho = 3.093299 \times 10^{-8}.$$

Exact reconstruction produced coefficients with denominator one.

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix},$$

the recovered algorithm computes

$$\begin{aligned} P_1 &= (b - d)h, & P_2 &= a(e + f + g + h), \\ P_3 &= (a - b)(g + h), & P_4 &= (a - c)f, \\ P_5 &= (c - d)e, & P_6 &= d(e + g), \\ P_7 &= (a - d)(e + g + h), \end{aligned}$$

and reconstructs

$$AB = \begin{pmatrix} -P_1 - P_3 + P_6 + P_7 & P_1 + P_2 - P_6 - P_7 \\ P_5 + P_6 & P_2 - P_4 - P_6 - P_7 \end{pmatrix}.$$

Expansion gives

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

Exact verification evaluated all

$$4 \times 4 \times 4 = 64$$

coefficients of the multiplication tensor. All coefficients matched, giving zero mismatches and maximum exact residual zero.

4 Discussion

The recovered identity attains the classical rank-7 bound for 2×2 matrix multiplication. Its coefficient representation differs from the standard presentation of Strassen’s construction, and it was recovered from random initial coefficients without access to Strassen’s formula.

The campaign also separates numerical approximation from mathematical verification. Seed 482 attained the lowest floating-point residual, while seed 581 produced the first exactly reconstructable identity. The numerical objective alone therefore does not determine whether a candidate admits the exact representation required for proof.

This experiment validates the numerical search, exact reconstruction, and independent verification pipeline. The same procedure can now be applied to finite (u, r) problems whose explicit decompositions are unknown.

Code availability

Source code, configurations, and complete run artefacts are available at github.com/switzerlandomics/ai-fast_mm_search.

References

- [1] V. Strassen, “Gaussian elimination is not optimal,” *Numerische Mathematik*, vol. 13, pp. 354–356, 1969. [doi:10.1007/BF02165411](https://doi.org/10.1007/BF02165411).
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